APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Fifth Semester B.Tech Degree Examination

MODEL QUESTION PAPER — SET 3

Course Code: PCCST501 **Course Name:** Computer Networks

Maximum Marks: 60 **Duration:** 2 hours 30 minutes

PART A — Answer all questions. Each question carries 3 marks.

Q. No.	Question	Marks	CO
1	Explain the role of DNS in translating a host name into information needed for Internet communication.	3	CO1
2	Distinguish between the World Wide Web and the Internet.	3	CO1
3	Explain how TCP flow control prevents a fast sender from overwhelming a receiver.	3	CO2
4	Describe the purpose of bind() in a TCP server program.	3	CO2
5	Explain the role of next-generation IP in supporting network-layer communication.	3	CO3
6	Distinguish between unicast routing and multicast routing in terms of delivery pattern.	3	CO3
7	Explain how the Ethernet protocol uses frames to support communication on a LAN.	3	CO4
8	Define transmission bandwidth and state its significance in data communication.	3	CO5

PART B — Answer one full question from each module. Each question carries 9 marks.

Module 1 — Question 9 (CO: CO1; Marks: 9)

(a) Explain the functions of HTTP and FTP and distinguish the kinds of application services they support. (4 marks)
(b) A user enters a domain name in a browser. Describe the sequence involving DNS resolution and an HTTP request that leads to retrieval of a web resource. (5 marks)

Module 1 — Question 10 (CO: CO1; Marks: 9)

(a) Explain how the peer-to-peer paradigm distributes file-sharing responsibilities among participating hosts. (4 marks)
(b) Illustrate the main stages by which BitTorrent peers obtain and share pieces of a file. (5 marks)

Module 2 — Question 11 (CO: CO2; Marks: 9)

(a) Explain the functions of TCP flow control, error control and congestion control. (5 marks)
(b) A TCP receiver expects byte 2401. It receives a segment carrying bytes 2601–2800 while bytes 2401–2600 are absent. State the cumulative acknowledgement number and justify it. (4 marks)

Module 2 — Question 12 (CO: CO3; Marks: 9)

(a) Explain how a TCP client/server application uses socket(), bind(), listen(), accept() and connect() to establish communication. (5 marks)
(b) A router has links with costs: A–B = 4, A–C = 2, C–B = 1, B–D = 5, C–D = 8. Compute the least-cost path from A to D and its total cost. (4 marks)

Module 3 — Question 13 (CO: CO4; Marks: 9)

(a) Explain the role of data-link control (DLC) in organizing communication over a link. (5 marks)
(b) Compare the functions of link-layer addressing and Ethernet framing. (4 marks)

Module 3 — Question 14 (CO: CO4; Marks: 9)

(a) Explain how wireless LANs provide link-layer communication and identify two characteristics that distinguish them from wired Ethernet. (5 marks)
(b) Describe how a switch uses destination link-layer addresses to forward frames in a LAN. (4 marks)

Module 4 — Question 15 (CO: CO5; Marks: 9)

(a) Explain the distinction between analog transmission and digital transmission with suitable signal descriptions. (5 marks)
(b) A communication link transfers 7,200 bits in 3 seconds. Compute the average data rate. (4 marks)

Module 4 — Question 16 (CO: CO5; Marks: 9)

(a) Explain how bandwidth utilization and transmission media influence achievable communication performance. (5 marks)
(b) A signal occupies a bandwidth of 4 kHz. Express this bandwidth in hertz and state what the unit indicates. (4 marks)

— End of Question Paper —

B. BLUEPRINT TABLE

Q. No.	Module	Syllabus topic	CO	Bloom's level	Marks
1	1	DNS; application-layer protocols	CO1	Understand	3
2	1	World Wide Web and Internet overview	CO1	Understand	3
3	2	TCP services: flow control	CO2	Understand	3
4	2	TCP sockets and server setup	CO2	Understand	3
5	2	Next-generation IP	CO3	Understand	3
6	2	Unicast and multicast routing	CO3	Understand	3
7	3	Ethernet protocol and frames	CO4	Understand	3
8	4	Bandwidth utilization; data and signals	CO5	Remember	3
9	1	HTTP, FTP, DNS	CO1	Apply	9
10	1	Peer-to-peer paradigm; BitTorrent	CO1	Apply	9
11	2	TCP flow/error/congestion control; sequence trace	CO2	Apply	9
12	2	TCP sockets; unicast routing path	CO3	Apply	9
13	3	Data-link control; link-layer addressing and framing	CO4	Understand/Analyse	9
14	3	Wireless LAN; connecting devices	CO4	Apply	9
15	4	Analog/digital transmission; data rate	CO5	Apply	9
16	4	Bandwidth utilization; transmission media	CO5	Apply	9

C. VERIFICATION CHECKLIST

Check	Result
Part A marks	8 × 3 = 24 marks.
Part B marks	Four answered questions × 9 = 36 marks.
Module distribution	Two Part A questions and two Part B alternatives per module.
Subdivision totals	Q9–Q16 each total 9 marks; each question has two subdivisions.
CO coverage	CO1, CO2, CO3, CO4 and CO5 are assessed.
Problem-solving	Q11 protocol trace; Q12 route-cost calculation; Q15 data-rate calculation.
Numerical verification	Q11 cumulative ACK = 2401. Q12 least path A–C–B–D costs 2+1+5 = 8 (less than A–B–D = 9 and A–C–D = 10). Q15: 7200/3 = 2400 bits/s. Q16: 4 kHz = 4000 Hz.
No repetition	New questions and scenarios relative to Sets 1 and 2; no question is copied or merely reworded.

Official Course Outcomes

CO1: Understand the internetworking design in terms of protocol stack and the role of various application layer protocols

CO2: Illustrate the functions of the transport layer from connectionless and connection-oriented perspectives

CO3: Identify how the network layer achieves host-to-host connectivity and caters to the diverse service requirements of the host applications

CO4: Explain the nuances of the data link layer design and demonstrate the various data link link layer protocols

CO5: Describe the fundamental characteristics of the physical layer and understand how the physical layer supports the functionalities of the top layers

D. SCHEME OF VALUATION

Part A

Q1: DNS resolves a name into relevant records/address information (2); enables applications to locate services/hosts (1).

Q2: WWW is a system of linked web resources/services (1.5); Internet is the underlying global network of interconnected networks (1.5).

Q3: Receiver advertises/communicates receive capacity (1); sender limits outstanding data accordingly (1); avoids receiver-buffer overflow (1).

Q4: bind() associates a socket with a local IP address and port (2); allows clients to reach a server at that endpoint (1).

Q5: Identifies next-generation IP as IPv6 (1); explains expanded address space (1); valid network-layer support benefit (1).

Q6: Unicast one destination (1); multicast group of destinations (1); valid delivery-pattern contrast (1).

Q7: Ethernet encapsulates network-layer data in frames (1); uses source/destination MAC addresses (1); supports LAN delivery (1).

Q8: Bandwidth is frequency range/channel capacity measure (1); expressed in hertz (1); relates to signal/data-carrying capacity (1).

Part B

Q9(a) — 4: HTTP transfers web resources (2); FTP supports file transfer (2).

Q9(b) — 5: DNS lookup (2); browser obtains destination address (1); HTTP request (1); response/resource retrieval (1).

Q10(a) — 4: Peers act as both consumers and providers (2); file pieces/resources are distributed among peers (2).

Q10(b) — 5: File divided into pieces (1); peers discover/connect to peers (1); download pieces (1); upload pieces to others (1); assemble complete file (1).

Q11(a) — 5: Flow control (2); error control/retransmission (2); congestion control (1).

Q11(b) — 4: Gap starts at byte 2401 (2); out-of-order later segment does not close gap (1); cumulative ACK remains 2401 (1).

Q12(a) — 5: socket() creates endpoint (1); bind() assigns local endpoint (1); listen() marks passive server (1); accept() accepts connection (1); connect() initiates client connection (1).

Q12(b) — 4: Path A–C–B–D (1); cost 2+1+5 (2); total cost 8 (1).

Q13(a) — 5: DLC organizes link-level data transfer (2); controls/structures frames or link operation (2); valid link-service role (1).

Q13(b) — 4: Link-layer address identifies local interface (2); Ethernet frame provides header/trailer encapsulation for delivery (2).

Q14(a) — 5: Wireless LAN transmits over radio (2); link-layer access coordination (1); two valid differences such as shared medium/interference/mobility (2).

Q14(b) — 4: Learns source address on ingress port (1); checks destination in forwarding table (1); forwards to matching port (1); floods unknown destination where applicable (1).

Q15(a) — 5: Analog signals vary continuously (2); digital signals use discrete levels/symbols (2); valid comparison of noise handling (1).

Q15(b) — 4: Rate = bits/time (1); 7200/3 (2); 2400 bits/s (1).

Q16(a) — 5: Bandwidth limits capacity (2); medium properties affect attenuation/noise (2); performance consequence (1).

Q16(b) — 4: 1 kHz = 1000 Hz (2); 4 kHz = 4000 Hz (1); hertz measures cycles per second/frequency range (1).